

Beyond Black-Box Assessment: Identifying Visual Properties that Predict Aesthetic Attraction in Urban Architecture

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Abstract

Understanding aesthetic attractiveness in built and natural environments has significant implications for urban planning, architectural design, landscape evaluation, and public wellbeing. Research shows that visually appealing places can influence how people experience, use, and value environments. However, aesthetic quality remains difficult to define objectively, making it challenging to establish evidence-based design guidelines. Identifying the specific visual properties that contribute to perceived attractiveness could provide architects, planners, and researchers with more interpretable criteria for evaluating scenic quality.

This study investigates which visual properties most strongly influence aesthetic judgment across urban and landscape images. Specifically, it examines whether Christopher Alexander's 15 fundamental properties of wholeness can predict human scenicness ratings, and whether quantifiable image features such as object count, edge density, and spatial distribution add explanatory value.

Preliminary analysis compared NIMA (Neural Image Assessment) scores with human ratings across 200 images (100 urban, 100 landscape). Although NIMA provides overall aesthetic scores, it showed important limitations: it compressed aesthetic judgments into a narrow range and offered no explanation of which visual properties influenced perception. This motivated a shift from black-box neural assessment to theory-driven property analysis.

The final methodology evaluated a dataset of 800 images, consisting of 400 urban and 400 landscape scenes, using Claude-based ratings of Alexander's 15 properties. The Segment Anything Model (SAM) was also used to quantify structural image features. Through correlation, regression, and feature-importance analysis, the study identified both shared and context-specific predictors of human scenicness ratings. Results indicate that Claude-based Alexander-property scores align meaningfully with human judgments, while SAM-derived features add modest predictive value, especially for landscape images. Overall, the study shows that aesthetic attraction can be partially explained through interpretable visual properties rather than treated only as a black-box score.

I. INTRODUCTION

Scenic quality is one of those qualities people recognize quickly but struggle to explain precisely. A street can feel inviting before anyone identifies why; a landscape can feel calm before its visual structure is consciously described. In this sense, aesthetic attraction is not merely a decorative concern. It shapes how people experience places, whether they choose to walk through them, linger in them, photograph them, invest in them, or avoid them. For cities, this matters directly: visual quality is tied to public wellbeing, neighborhood perception, economic value, and the everyday emotional experience of the built environment. For natural and semi-natural landscapes, it matters as well, because scenicness often influences conservation priorities, tourism, recreation, and broader judgments about environmental quality. Recent computational and crowd-sourced studies have shown that people can make surprisingly consistent judgments about the attractiveness, safety, and beauty of places, even when those judgments are collected across large and diverse populations (Salesses et al., 2013; Dubey et al., 2016; Seresinhe et

al., 2017). This consistency suggests that aesthetic perception is not simply random personal taste. Something in the visual structure of places appears to be shared, recognizable, and measurable.

The ideal situation, especially for architecture, planning, and environmental design, would be to move from vague aesthetic language to evidence-based visual understanding. Designers should not only be able to say that one street, building frontage, or landscape view is more attractive than another; they should be able to identify which visual properties create that attraction. Is the appeal driven by a clear focal structure? By gradual transitions? By contrast, positive space, simplicity, coherence, or multi-scale detail? These are not minor questions. Aesthetic judgments influence design guidelines, urban regeneration, housing desirability, public-space use, and the perceived identity of neighborhoods. Yet in practice, aesthetic quality is still often treated as subjective, intuitive, or too holistic to decompose. As a result, designers and planners may know that visual attractiveness matters, but they lack a reliable way to translate human perception into measurable and actionable design features.

This gap becomes more visible when we examine current computational approaches to image-based aesthetic assessment. Deep learning models such as Neural Image Assessment, or NIMA, have made it possible to assign aesthetic scores to images using convolutional neural networks trained on large human-rated datasets. NIMA builds on the broader tradition of aesthetic visual analysis, including datasets such as AVA, which contains more than 250,000 images with human aesthetic ratings (Murray et al., 2012; Talebi & Milanfar, 2018). These models are useful because they show that aesthetic judgment can be approximated computationally. However, they also reveal a central limitation: they predict without explaining. A single score may indicate that an image is likely to be attractive, but it does not tell us which visible properties caused that judgment. It cannot tell an architect whether a façade needs stronger centers, better boundaries, richer gradients, less clutter, or more visual unity.

This limitation was also evident in the preliminary phase of the present project. NIMA was tested as a black-box baseline against human scenicness ratings, but it compressed aesthetic judgments into a narrow range. In the project draft, NIMA scores had a standard deviation of approximately 0.246, compared with 1.598 for human ratings, suggesting that the model under-discriminated the variation humans actually perceive. More importantly, NIMA did not provide any explanation of which visual properties were responsible for the scores. The project presentation similarly reports that NIMA produced a limited score range of roughly 4.6–6.0 compared with the full 1–10 spread in human ratings, motivating the shift toward an interpretable, property-based approach.

A more interpretable foundation can be found in Christopher Alexander's theory of wholeness. In *The Nature of Order*, Alexander (2002) proposed 15 fundamental properties that appear repeatedly in environments perceived as coherent, beautiful, and "alive." These properties include levels of scale, strong centers, boundaries, alternating repetition, positive space, good shape, local symmetries, deep interlock and ambiguity, contrast, gradients, roughness, echoes, the void, simplicity and inner calm, and not-separateness. What makes this framework especially relevant is that it does not describe beauty as surface decoration. Instead, it describes relationships among parts: how elements support one another, how spaces are shaped, how repetition avoids monotony, how contrast gives structure, and how a scene becomes visually unified rather than fragmented. In other words, Alexander's framework offers a vocabulary for decomposing aesthetic attraction into design-relevant properties.

Even so, Alexander's theory remains under-tested in large-scale empirical settings. It is influential in architectural theory, design philosophy, and pattern-based thinking, but its 15 properties are rarely operationalized as measurable variables across large image datasets. This is where the current research creates its space. The issue is not simply whether Alexander's theory is conceptually persuasive; the stronger question is whether these properties actually predict human ratings of scenicness and aesthetic attraction. Do people rate images higher when those images contain stronger centers, clearer positive

space, richer gradients, or greater not-separateness? Are the same properties important in urban scenes and landscape scenes, or do built and natural environments rely on different visual cues? These questions remain only partly answered in existing research.

Recent advances in computer vision and multimodal AI make this question newly testable. Large language models with visual reasoning capabilities can be prompted to evaluate theory-driven visual properties consistently across image sets. At the same time, segmentation models such as the Segment Anything Model, or SAM, allow researchers to extract structural image features such as segment count, object coverage, edge density, compactness, and spatial distribution. SAM is especially useful because it was designed as a promptable segmentation model that can generalize across image types and segmentation tasks (Kirillov et al., 2023). This creates an opportunity to combine two kinds of evidence: theory-driven perceptual ratings from Alexander's framework and lower-level structural features derived from segmentation.

The present study therefore moves beyond black-box aesthetic scoring by asking which visual properties most strongly predict human scenicness ratings across urban and landscape images. Although the title emphasizes urban architecture, the study uses both built and natural scenes because landscape images provide an important comparative context. This matters because aesthetic attraction may not operate in the same way across environments. An urban street may depend more heavily on boundaries, centers, positive space, and simplicity, while a landscape may depend more strongly on gradients, openness, good shape, or visual continuity. Treating landscape images as a comparison group helps clarify which visual properties are general predictors of scenicness and which are context specific.

The empirical design reflects this goal. The project uses 800 images from Geograph and Scenic-or-Not, balanced across 400 urban and 400 landscape images. Human scenicness ratings are drawn from more than 212,000 crowd-sourced votes, providing a ground-truth measure of perceived aesthetic quality on a 1–10 scale. Each image is evaluated using Claude on Alexander's 15 properties, producing 12,000 theory-driven property ratings. The project also includes SAM-based structural features and compares model performance across feature sets. According to the project presentation, Claude-based property scores show meaningful agreement with human ratings, with correlations reaching approximately $r = 0.67$ overall for gradients, $r = 0.69$ in landscape images for good shape, and $r = 0.62$ in urban images for simplicity and inner calm. SAM features add modest predictive value, especially for landscape images, but the strongest drivers remain largely Alexander-property scores rather than purely structural segmentation features.

The study is guided by the following objectives. First, it tests whether Christopher Alexander's 15 fundamental properties can be operationalized as measurable image-level predictors of human scenicness ratings. Second, it evaluates which of these properties contribute most strongly to perceived aesthetic attraction across all images. Third, it compares urban and landscape scenes to identify whether aesthetic drivers are universal or environment specific. Fourth, it examines whether SAM-derived structural features add predictive value beyond theory-driven property ratings. Finally, it uses NIMA as a baseline example of black-box aesthetic assessment to show why interpretability is necessary for design-relevant explanation.

The contribution of this research is therefore both methodological and practical. Academically, it connects urban perception research, computational aesthetics, environmental scenicness studies, and Alexander's architectural theory of wholeness. Methodologically, it demonstrates how large language models and image segmentation can be used together to translate qualitative design theory into quantitative analysis. Practically, it offers a way to move from "this place looks attractive" to "these visual properties appear to drive attractiveness." That shift is small in wording but large in implication. It

can support more interpretable design evaluation, evidence-based planning, and clearer communication between researchers, designers, and policymakers.

II. BACKGROUND AND RELATED WORK

Scenic quality and aesthetic attraction are central but difficult-to-measure dimensions of how people perceive places. In both urban and landscape environments, visual attractiveness affects more than personal preference: it influences how people evaluate neighborhoods, choose routes, attach meaning to places, and support design or planning interventions. The present study, *Beyond Black-Box Assessment: Identifying Visual Properties that Predict Aesthetic Attraction in Urban Architecture*, is situated within this broader effort to understand whether aesthetic perception can be decomposed into measurable visual properties. Its objectives are to test whether Christopher Alexander's 15 fundamental properties of wholeness predict human scenicness ratings, to compare these relationships across urban and landscape images, to evaluate the limitations of black-box aesthetic assessment using NIMA, and to examine whether SAM-derived structural image features add explanatory value beyond theory-driven property ratings.

A. Scenicness and Human Aesthetic Perception

A major starting point for this study is the evidence that human judgments of place-based attractiveness are more consistent than commonly assumed. Although scenicness and aesthetic appeal are often described as subjective, large-scale perceptual studies suggest that people show meaningful agreement when evaluating visual environments. Salesses, Schechtner, and Hidalgo (2013), through the Place Pulse project, demonstrated that crowdsourced image comparisons could reveal shared perceptions of urban safety, class, and attractiveness. Their contribution was important because it translated subjective impressions of cities into mappable and quantitative data. However, the study mainly focused on perceived outcomes rather than the visual mechanisms behind those outcomes. It showed that people often agree about how places feel, but it did not fully explain which features of buildings, streets, open spaces, or landscapes shaped those judgments.

Dubey, Naik, Parikh, Raskar, and Hidalgo (2016) extended this work by using deep learning to quantify urban perception at a global scale. Their research demonstrated that machine learning models could predict perceptual qualities of city images across geographically diverse contexts. This was methodologically significant because it established that urban perception could be modeled computationally from large image datasets. Yet the limitation remained interpretability. A model may predict whether an image appears safe, lively, or attractive, but it does not necessarily explain whether that judgment comes from scale, contrast, boundaries, spatial order, greenery, openness, or visual coherence. For planners and designers, this distinction matters. Prediction alone is not enough if the goal is to understand how attractive environments can be intentionally created.

Landscape-focused research has reached similar conclusions. Seresinhe, Preis, and Moat (2017) used deep learning to quantify the beauty of outdoor places, showing that scenicness can be predicted from large-scale image data. Their study is especially relevant because it treats scenic beauty as a measurable environmental variable rather than as an entirely private emotional response. It also broadens the discussion beyond urban streets to include natural and semi-natural environments. However, like much deep learning research, it emphasized predictive performance more than theoretical explanation. It showed that images could be scored for beauty, but it did not provide a detailed account of which compositional or structural properties caused higher ratings. This gap directly motivates the present study, which aims not only to predict scenicness but also to identify the visual properties associated with human aesthetic judgments across both urban and landscape images (Seresinhe et al., 2017).

B. Computational Aesthetic Assessment and NIMA

Computational aesthetic assessment models represent another important foundation for this research. Among these, Neural Image Assessment, or NIMA, is especially relevant. Talebi and Milanfar (2018) proposed NIMA as a convolutional neural network model trained to predict human aesthetic ratings. Rather than producing only a simple binary classification of good or bad image quality, NIMA predicts a distribution of possible human ratings and calculates an expected aesthetic score. This makes it more nuanced than many earlier image-quality assessment models and explains why it is often useful as a baseline in computational aesthetics.

However, NIMA also illustrates the core limitation of black-box aesthetic assessment. It was developed primarily for general photographic aesthetics, not for explaining the visual structure of places. Its output is a scalar or expected aesthetic score; it does not reveal whether beauty arises from strong centers, gradual transitions, openness, contrast, positive space, symmetry, or simplicity. For architecture, landscape evaluation, and urban planning, this is a serious limitation. A black-box score may classify an image as attractive, but it cannot indicate which design properties should be preserved, modified, or reproduced.

This limitation is directly connected to the present research. The purpose of using NIMA is not to treat it as the final explanatory model, but to position it as a baseline example of the predictive-but-opaque approach. By contrast, this study seeks to move toward interpretable aesthetic modeling. The central question is not only whether a model can approximate human scenicness ratings, but whether it can help explain which visual properties contribute to those ratings.

C. Christopher Alexander’s 15 Properties of Wholeness

The theoretical counterpoint to black-box scoring is Christopher Alexander’s theory of wholeness. In *The Nature of Order*, Alexander (2002) argued that environments perceived as beautiful, coherent, or “alive” tend to contain 15 recurring structural properties: levels of scale, strong centers, boundaries, alternating repetition, positive space, good shape, local symmetries, deep interlock and ambiguity, contrast, gradients, roughness, echoes, the void, simplicity and inner calm, and not-separateness. The value of this framework lies in its interpretability. Unlike purely statistical image features, Alexander’s properties are directly meaningful for design analysis. They describe how parts relate to wholes, how visual order emerges, and how complexity can remain coherent rather than chaotic.

Alexander’s framework is particularly useful for this study because it offers a vocabulary for decomposing aesthetic attraction into recognizable visual components. For example, “levels of scale” refers to the presence of multiple nested sizes, while “strong centers” describes focal elements that organize the visual field. “Positive space” refers to well-shaped spatial regions, and “not-separateness” describes the sense that elements belong to their surroundings rather than appearing isolated. These concepts are closer to the language of architecture and design than conventional low-level image features such as brightness, color histogram, or edge count.

At the same time, Alexander’s theory has often remained theoretical or descriptive. Its claims are influential, but they have not been sufficiently tested at scale against human scenicness ratings. Research inspired by Alexander, such as Jiang and Sui (2013), has argued that living structure and wholeness can be examined through computational and geographic perspectives. This work is valuable because it reframes Alexander’s theory as something that may be measurable rather than purely philosophical. Still, concepts such as “not-separateness” or “inner calm” are abstract and difficult to capture with traditional computer vision metrics alone. This creates a methodological opening for multimodal AI, especially large language models capable of structured visual reasoning. By using Claude to score images according to

Alexander’s 15 properties, the present study attempts to bridge qualitative architectural theory and quantitative empirical testing.

D. Computer Vision and SAM-Based Structural Features

Recent advances in computer vision provide a complementary route to interpretable scenicness analysis. The Segment Anything Model, introduced by Kirillov et al. (2023), enables scalable segmentation across diverse image types. This makes it possible to extract structural features such as object count, segment coverage, edge density, compactness, symmetry, and spatial distribution. These features are not equivalent to Alexander’s properties, but they describe the physical organization of an image in measurable terms.

SAM-based features are useful because they provide structural evidence independent of language-model ratings. For example, segment count may indicate object richness or visual clutter, while edge density may reflect textural complexity. Coverage and compactness may help describe whether image elements are concentrated, fragmented, or spatially balanced. These measures can support the analysis of how visual organization relates to scenicness.

However, segmentation features alone may still be too low-level to explain aesthetic perception fully. A scene can contain many objects without being beautiful, and a visually simple scene can feel either calm or empty depending on its organization. Similarly, high edge density might represent rich visual detail in one image but clutter in another. This suggests that SAM features are best treated as complementary structural predictors rather than replacements for theory-driven visual properties. In the present study, SAM-derived features are used to examine whether measurable spatial and compositional characteristics add explanatory value beyond Claude-based ratings of Alexander’s properties.

E. Research Gap

Across the literature, a clear pattern emerges: human judgments of scenicness and aesthetic attraction are more consistent than expected, and computational models can predict many of these judgments at scale. However, prediction has advanced faster than explanation. Place Pulse-style studies quantify perception but do not fully decompose its visual causes. Deep learning models such as NIMA and scenicness predictors produce useful scores, but they remain limited by opacity. Alexander’s theory provides a rich explanatory vocabulary, yet it lacks large-scale empirical validation. SAM and related segmentation models offer measurable structural features, but they may not fully capture higher-order perceptual qualities such as coherence, calm, and wholeness.

The existing literature therefore only partially addresses the aims of the present study. Prior research supports the idea that aesthetic judgments can be measured and modeled, but it does not yet provide a sufficiently interpretable account of which visual properties drive those judgments across urban and landscape environments. This study addresses that gap by integrating human scenicness ratings, NIMA baseline comparison, Claude-based evaluation of Alexander’s 15 properties, and SAM-derived structural features. In doing so, it contributes a more interpretable framework for understanding aesthetic attraction: one that does not stop at predicting whether a place is beautiful, but asks which properties make it so.

III. METHODOLOGY

A. Dataset and Image Collection

This study used images obtained from the Geograph UK dataset, a large collection of geographically referenced photographs of built and natural environments. Images were downloaded programmatically using the Geograph API. A balanced sample of 800 images was selected for analysis, consisting of 400

urban images and 400 landscape images. This structure allowed the study to compare whether the visual properties associated with aesthetic attraction differ between built environments and natural landscapes.

Human scenicness ratings were obtained from the Scenic-Or-Not dataset, which provides crowd-sourced evaluations of image beauty on a 1–10 scale. These ratings were used as the ground-truth measure of perceived aesthetic quality. In total, the dataset included more than 212,000 human votes, allowing each image to be linked to an aggregate human scenicness score.

B. Claude Baseline Model: Alexander’s 15-Property Evaluation

The baseline model in this study was a Claude-based property scoring approach. Each image was evaluated using the Claude API according to Christopher Alexander’s 15 fundamental properties of wholeness. These properties include levels of scale, strong centers, boundaries, alternating repetition, positive space, good shape, local symmetries, deep interlock and ambiguity, contrast, gradients, roughness, echoes, the void, simplicity and inner calm, and not-separateness.

For each image, Claude assigned a score from 1 to 10 for every property. Since the dataset contained 800 images and each image was rated across 15 properties, this process generated 12,000 property-level ratings. These Claude-generated scores were treated as interpretable visual features and used to examine which Alexander properties most strongly predict human scenicness ratings.

C. SAM-Based Structural Features

To complement the Claude-based property ratings, the Segment Anything Model was used to extract structural features from each image (Kirillov et al., 2023). SAM-based analysis produced 14 image-level features, including segment count, edge density, compactness, symmetry, coverage, gradient-related measures, and other spatial characteristics. These features were included to test whether measurable structural properties of images add predictive value beyond the Claude-based Alexander-property scores.

D. Feature Importance and Statistical Analysis

The main objective of the analysis was to identify which visual properties contribute most strongly to human aesthetic perception. Human scenicness ratings were used as the target variable, while Claude-based Alexander-property scores and SAM-derived features were used as predictors.

Feature importance was evaluated using a five-method ensemble consisting of Lasso regression, Ridge regression, Random Forest, permutation importance, and Gradient Boosting (Tibshirani, 1996; Hoerl & Kennard, 1970; Breiman, 2001; Friedman, 2001; Pedregosa et al., 2011). Results were analyzed for the full dataset and separately for urban and landscape images. This allowed the study to determine whether the same visual properties predict aesthetic attraction across both categories or whether urban and landscape scenes depend on different aesthetic drivers.

IV. RESULTS

A. Score Distribution Between Human Ratings and Claude Scores

The first stage of analysis compared the distribution of human scenicness ratings with the Claude-generated aesthetic property scores. Human ratings showed a broader spread across the 1 to 10 scale, while Claude ratings were more concentrated and generally higher. The mean human scenicness score

was 4.72, compared with a Claude mean score of 6.62. This indicates that Claude tended to assign more favorable ratings than the human crowd-sourced judgments. The difference was also visible in the standard deviation: human ratings had a standard deviation of 1.78, whereas Claude ratings had a lower standard deviation of 1.21. These values suggest that human judgments contained greater variation, while Claude produced a narrower and more optimistic score distribution.



Figure 1. Distribution of human scenicness ratings and Claude mean scores. Human ratings show a wider spread across the 1–10 scale, while Claude scores are more concentrated and generally higher. This indicates that Claude produced a narrower and more optimistic score distribution than the human crowd-sourced ratings.

This difference is important because it shows that Claude did not reproduce human ratings directly. Instead, it provided a structured interpretive assessment of each image according to Alexander’s visual properties. The observed positive scoring bias should therefore be considered when interpreting the results. Claude’s role in this study was not to replace human judgment, but to generate consistent property-level ratings that could be tested against human scenicness scores. The higher average rating suggests that Claude may recognize aesthetic structure more generously than human raters, but the correlation results below show that the relative ordering of visual properties still captured meaningful relationships with human perception.

B. Agreement Between Claude Property Scores and Human Scenicness Ratings

The correlation analysis showed that several Claude-rated Alexander properties were meaningfully associated with human scenicness ratings. Across the full dataset, Gradients emerged as the strongest single property, with a correlation of $r = 0.67$. This indicates that images with smoother transitions, gradual visual change, and layered shifts in form, light, texture, or spatial depth tended to receive higher human scenicness ratings. The strength of this relationship suggests that gradual visual organization may be one of the most generalizable features of aesthetic attraction across both urban and landscape environments.

When the analysis was separated by image type, different properties emerged as the strongest predictors. For landscape images, Good Shape showed the strongest relationship with human ratings, reaching $r = 0.69$. This suggests that landscapes perceived as more aesthetically pleasing were often those with coherent, well-formed visual structures rather than fragmented or visually awkward compositions. For urban images, Simplicity and Inner Calm was the strongest driver, with $r = 0.62$. This finding suggests that attractive urban scenes were not necessarily the most visually complex or dense, but those that conveyed order, clarity, and reduced clutter.

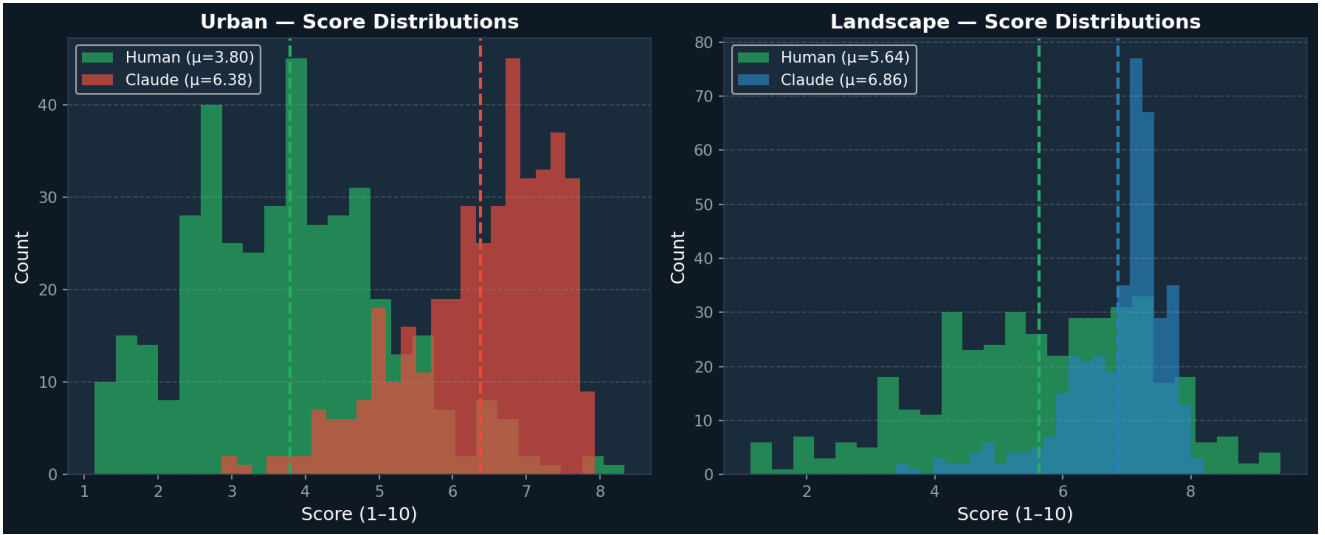


Figure 2. Claude mean score versus human scenicness rating for urban and landscape images. Scatterplots show the relationship between Claude’s average Alexander-property score and crowd-sourced human scenicness ratings. The dashed regression lines indicate that higher Claude mean scores generally correspond to higher human ratings in both image categories, with stronger agreement for landscape images.

Together, these results support the central assumption of the study: Alexander’s properties can be operationalized as measurable image-level predictors that relate meaningfully to human aesthetic judgments. The correlations also suggest that aesthetic attraction is not driven by one universal property alone. Instead, the visual properties that matter most vary depending on whether the image represents a built urban environment or a natural landscape.

C. Feature Importance Across Urban and Landscape Images

The feature-importance analysis provided a more detailed view of which variables contributed most strongly to human scenicness ratings. For urban images, all of the top five predictors were Claude-rated Alexander properties. The strongest urban driver was Simplicity and Inner Calm with a reported correlation of $r = +0.591$, followed by Positive Space with $r = +0.618$, Not-Separateness with $r = +0.605$, Levels of Scale with $r = +0.584$, and Gradients with $r = +0.529$. Although the ordering came from the composite feature-importance procedure rather than correlation alone, the consistently positive correlations indicate that these properties were meaningfully aligned with human ratings.

This pattern suggests that urban attractiveness was strongly associated with visual coherence and spatial organization. Simplicity and Inner Calm points to the importance of clarity and reduced visual noise. Positive Space indicates that the shape and organization of spaces within the image mattered, not only the objects themselves. Not-Separateness suggests that attractive urban scenes were those in which buildings, streets, vegetation, and surrounding elements appeared visually integrated. Levels of Scale further indicates that images with multiple coordinated scales, from larger spatial structure to finer detail, were rated more favorably.

For landscape images, the top predictors showed a slightly different structure. Positive Space ranked first with $r = +0.620$, followed by Good Shape with $r = +0.692$, Levels of Scale with $r = +0.652$, Edge Density with $r = -0.108$, and Simplicity and Inner Calm with $r = +0.600$. Unlike the urban model, the landscape top five included one SAM-derived structural feature: Edge Density. Its negative correlation suggests that higher edge density may have been associated with lower scenicness in landscape scenes, possibly because excessive edge complexity can indicate visual clutter or fragmentation rather than calm, coherent scenic structure.

The difference between the urban and landscape rankings is one of the most important findings of the study. Urban images were best explained entirely by Claude-rated Alexander properties, while landscape images benefited slightly from the inclusion of a structural computer-vision feature. This suggests that theory-driven properties were especially strong for built environments, whereas landscape perception may also depend on measurable visual texture and edge structure.

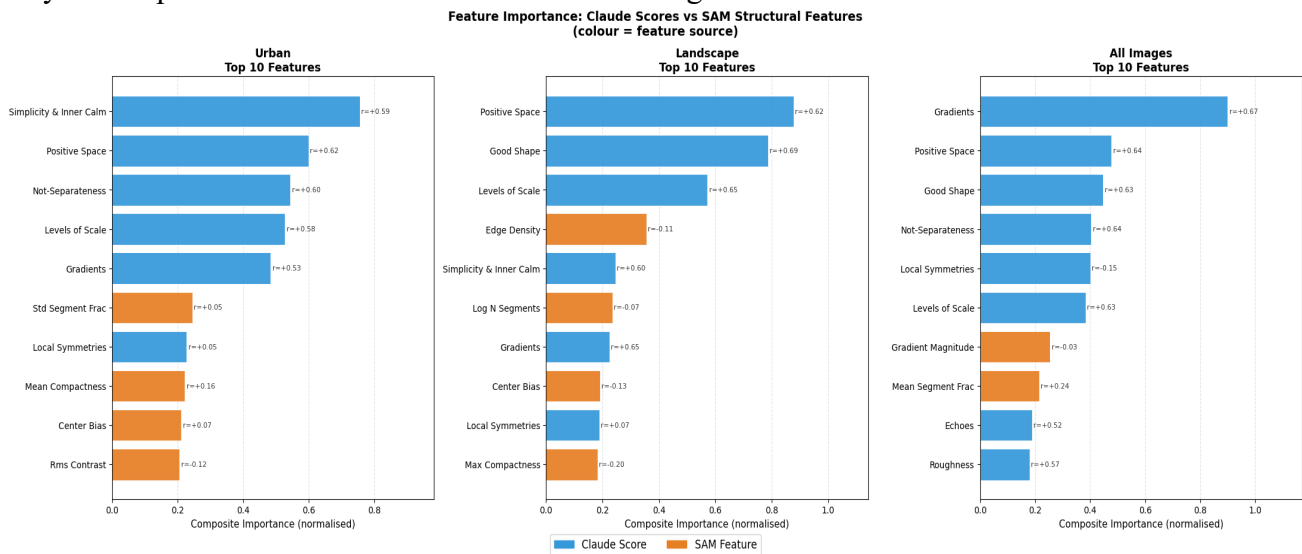


Figure 3. Top visual predictors of human scenicness by image type. The figure compares the highest-ranked predictors for urban and landscape images. Urban ratings were primarily associated with Claude-rated Alexander properties such as Simplicity and Inner Calm, Positive Space, Not-Separateness, Levels of Scale, and Gradients. Landscape ratings were most strongly associated with Positive Space, Good Shape, Levels of Scale, Edge Density, and Simplicity and Inner Calm.

D. Contribution of SAM-Derived Structural Features

The analysis also examined whether SAM-derived features improved predictive performance beyond Claude-based Alexander-property ratings. Across all images, adding SAM features increased the model R^2 from 0.468 to 0.487, a gain of +0.019. For urban images, R^2 increased from 0.394 to 0.427, a gain of +0.033. For landscape images, the improvement was larger, with R^2 increasing from 0.518 to 0.576, a gain of +0.058.

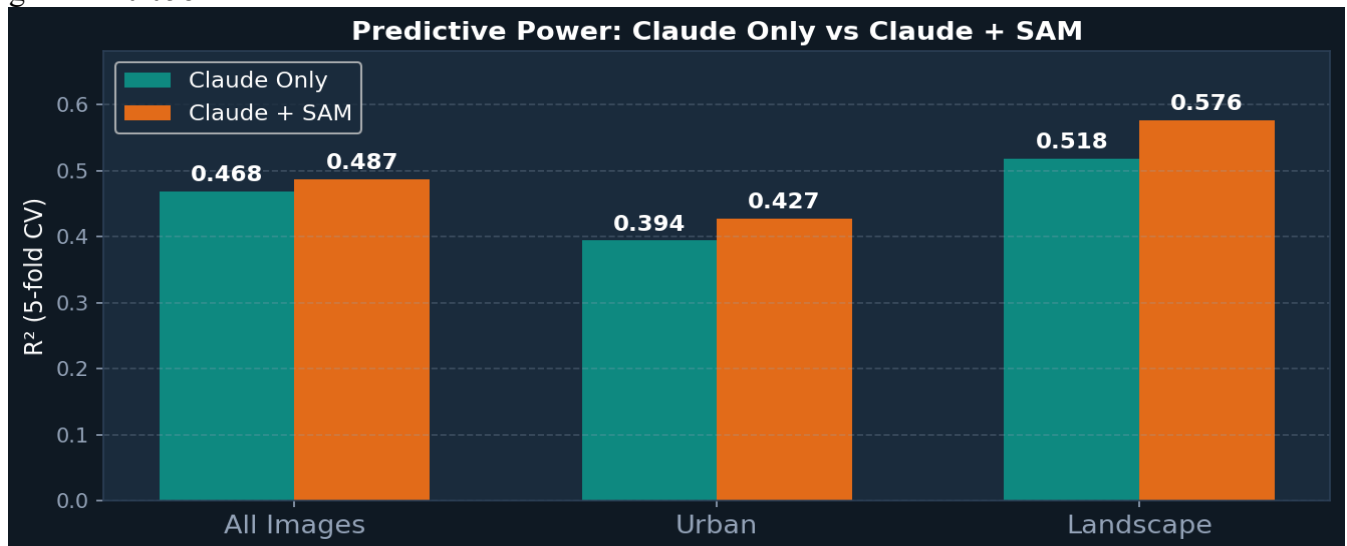


Figure 4. Predictive performance with and without SAM-derived structural features. The figure compares model R^2 before and after adding SAM features. SAM produced modest improvement overall, with the largest gain for landscape images, suggesting that segmentation-derived structure adds useful but secondary information beyond Claude-based Alexander-property scores.

These results show that SAM features added predictive value, but the improvement was modest. The gain was smallest for the full dataset and urban subset, and moderate for landscape images. This pattern supports the interpretation that SAM captures some real structural information, particularly in landscape scenes, but does not replace the explanatory value of Claude-rated Alexander properties. The project presentation also notes that Edge Density and Gradient Magnitude correlated most strongly with Claude’s perceptual scores, while most SAM features showed relatively weak correlation, explaining their lower importance ranking.

The limited contribution of SAM is meaningful rather than disappointing. It suggests that low-level structural features such as segments, edges, compactness, and coverage can describe parts of image organization, but they may not fully capture higher-order aesthetic qualities such as calmness, coherence, wholeness, or visual belonging. Those qualities appear to be better represented by the Alexander-property ratings generated through Claude.

E. Summary of Key Findings

Overall, the results indicate that Claude-based ratings of Alexander’s 15 properties were meaningfully associated with human scenicness judgments. The strongest overall relationship was found for Gradients, while the strongest category-specific properties differed between urban and landscape images. Urban attractiveness was most closely associated with Simplicity and Inner Calm, Positive Space, Not-Separateness, Levels of Scale, and Gradients. Landscape attractiveness was most closely associated with Positive Space, Good Shape, Levels of Scale, Edge Density, and Simplicity and Inner Calm.

The results also show that SAM-derived structural features provided only limited additional explanatory power. Their contribution was most visible in landscape images, where Edge Density entered the top-five feature ranking and the R^2 gain was larger than in the urban subset. However, across the study, Claude-based Alexander properties dominated the prediction of human scenicness ratings. This supports the main argument of the project: interpretable, theory-driven property assessment offers a more useful explanation of aesthetic attraction than black-box scoring alone.

V. DISCUSSION

A. Interpreting the central pattern

The findings indicate that the study’s theory-driven approach captured a meaningful portion of human aesthetic judgment while also revealing clear limits in the proxy measures used to operationalize it. Most importantly, Claude-derived property scores were moderately but consistently aligned with crowd-sourced scenicness ratings, with stronger agreement for landscapes than for urban scenes. The manuscript reports that Claude’s overall mean score was 6.62 compared with a human mean of 4.72, indicating a systematically more optimistic and compressed distribution; the same pattern appeared within subsets, where urban means were 6.38 for Claude versus 3.80 for humans, and landscape means were 6.86 versus 5.64. Even so, the relationship between Claude’s average property score and human judgments remained substantial, with Pearson correlations of 0.609 for urban images and 0.687 for landscape images, both with very small p-values.

The more analytically consequential result is that aesthetic attraction was not explained by an undifferentiated global score, but by a relatively coherent bundle of visual properties. Across the full dataset, Gradients was the strongest single correlate of scenicness ($r = 0.67$). In urban scenes, the most important predictors clustered around Simplicity and Inner Calm, Positive Space, Not-Separateness, Levels of Scale, and Gradients; in landscapes, Positive Space, Good Shape, Levels of Scale, and Simplicity and Inner Calm remained central, while Edge Density entered the top rank with a negative association. SAM-derived structural features modestly improved predictive power, but only at the

margin: R^2 rose from 0.468 to 0.487 for all images, from 0.394 to 0.427 for urban scenes, and from 0.518 to 0.576 for landscapes. Taken together, these results suggest that human scenicness judgments are more strongly connected to higher-order spatial organization than to purely low-level segmentation statistics.

B. Relating the findings to earlier research

These results are broadly consistent with prior work showing that people make surprisingly stable judgments about the visual quality of places (Salesses et al., 2013; Dubey et al., 2016; Seresinhe et al., 2017) and that such judgments can be modeled computationally. Studies of urban perception demonstrated that crowd ratings extracted from images can be aggregated into reproducible measures of how places are perceived, while later work scaled this logic globally using large street-view datasets and neural ranking models. Likewise, scenicness research in Great Britain showed that beauty is not reducible to raw naturalness (Seresinhe et al., 2017; Levering et al., 2021): some built elements can increase scenic ratings, whereas visually bland greenery does not necessarily do so. The present findings agree with that literature in two important ways. First, they support the claim that aesthetic judgments are measurable at scale. Second, they reinforce the idea that what matters is not simply whether a scene is urban or natural, but how its visual parts are organized.

At the same time, the findings sharpen and partially revise earlier environmental-preference research. Classic work in environmental psychology reported that well-structured urban spaces tend to be preferred and that coherence is a reliable positive predictor of urban preference; related meta-analytic work also framed coherence, complexity, legibility, and mystery as recurring, though heterogeneous, predictors of visual preference. The present study aligns most strongly with the coherence side of that literature. Positive Space, Not-Separateness, Good Shape, and Levels of Scale all point toward compositional unity rather than mere visual stimulation. Yet there is also a productive tension with earlier claims that complexity tends to raise preference. Here, raw Edge Density was negatively associated with scenicness in landscapes, and SAM features were generally weaker than the Alexander-property scores. The most plausible explanation is that the present study distinguishes between organized complexity and undifferentiated visual clutter. In other words, what seems to matter is not complexity in itself, but complexity that remains legible, nested, and calm.

C. Theoretical significance

Theoretically, the findings offer substantive support for Alexander's theory of wholeness. Alexander proposed that environments judged as beautiful or "alive" share recurrent structural properties and that these properties are, in principle, measurable. The present results do not prove the full theory, but they do provide empirical evidence that several of the fifteen properties covary with human scenicness in systematic ways. That is especially important because Alexander's framework has often been treated as philosophically suggestive but empirically difficult to test. By operationalizing the properties at scale and comparing them against crowd judgments, this study moves the theory from descriptive architectural discourse toward testable explanatory modeling.

The findings also have broader theoretical implications beyond Alexander. They fit well with psychophysical accounts of landscape quality, which argue that aesthetic value emerges from the interaction between measurable environmental attributes and human perceptual judgment. They also resonate with informational models of preference in which coherence and legibility support immediate understanding, while complexity and mystery support exploration. What this study adds is a more architecture-centered vocabulary for those same mechanisms. Simplicity and Inner Calm, Positive Space, Good Shape, and Levels of Scale appear to be architectural counterparts of coherence and

legibility. This is a novel contribution because it suggests that the gap between environmental psychology and architectural theory may be narrower than usually assumed.

D. *Implications for theory, policy, and practice*

The practical and policy implications are significant, but they should be framed carefully. For theory, the main implication is that interpretable aesthetics models can do something black-box scores cannot: they can indicate which visual properties are likely driving preference. This matters because systems such as AVA-based aesthetics models and NIMA are useful predictors, yet their outputs are difficult to translate into actionable design knowledge; by contrast, a property-based model can tell planners and designers whether a scene is lacking calm, positive space, scale hierarchy, or visual integration. For policy, that means urban design review, heritage conservation, streetscape regulation, and public-realm investment could be informed by measurable visual criteria rather than relying solely on expert intuition or ideological style preferences. For professional practice, the results imply that designers should not equate attractiveness with maximum novelty, density, or object richness. Rather, the strongest signal across the findings is that preferred environments balance richness with formal coherence. Still, these results should guide judgment, not replace it, because the urban–landscape differences in the models show that aesthetic value remains contextual rather than universal in a simplistic sense.

E. *Limitations*

Several limitations temper the interpretation of the results. First, the study’s most innovative measure the Claude rating of Alexander’s properties is also a source of uncertainty. The manuscript itself shows that Claude produced a narrower and more favorable score distribution than human raters, which means the model likely compressed disagreement and may have muted weaker relationships. Second, the study relies on static images from a single national context and on crowd-sourced scenicness ratings drawn from photographed scenes, so the results may reflect not only environmental properties but also photographer choices, framing, weather, and context-specific aesthetic norms. Third, the operationalization of the Alexander properties remains indirect: an LLM is interpreting a theoretical vocabulary on behalf of the researcher, rather than those properties being coded independently by human raters or measured geometrically. That matters because prior meta-analytic work has shown that results can vary depending on whether the same population rates both preference and predictor variables. Fourth, the limited added value of SAM may say less about the irrelevance of structure than about the coarseness of the current structural proxies, which were not designed specifically for architectural morphology or experiential urban space. Finally, the analysis is correlational and image-based, so it cannot establish that changing one measured property in the real world would causally increase scenicness.

F. *Future research*

Future research should now move in five precise directions. First, Alexander-property ratings should be validated through multi-rater designs that compare LLM scores with trained human coders and with alternative models, reporting inter-rater reliability and calibration error. Second, the framework should be tested across multiple cultural and geographic contexts, especially outside Great Britain, to determine which predictors generalize and which are place-specific. Third, researchers should use causal designs such as controlled image editing, generative design manipulations, or virtual-environment experiments to test whether increasing Positive Space, Levels of Scale, or Simplicity and Inner Calm actually raises aesthetic ratings. Fourth, future models should incorporate richer spatial descriptors, including depth, façade parsing, isovists, graph-based measures of configuration, and architecture-specific segmentation, rather than relying on generic object masks alone. Fifth, the explanatory framework should be extended beyond scenicness to outcomes that matter socially and politically, such as restoration, walkability,

neighborhood attachment, and public-health benefits. Those steps would preserve the present study's central achievement making aesthetics more interpretable while addressing its current limits in validity, causality, and generalizability.

VI. CONCLUSION

This study set out to move beyond black-box aesthetic assessment by identifying which visual properties most strongly predict human perceptions of scenicness and aesthetic attraction across urban and landscape images. Using Christopher Alexander's 15 properties of wholeness as the theoretical foundation, the study combined Claude-based property scoring, crowd-sourced Scenic-Or-Not human ratings, and SAM-derived structural features to examine whether aesthetic judgment could be decomposed into interpretable visual components rather than treated as a single opaque score. The central objective was not only to predict which images people found more attractive, but to understand which visual qualities appeared to support those judgments.

The findings show that Claude-based ratings of Alexander's properties were meaningfully aligned with human scenicness ratings, although Claude scores were generally higher and less variable than human scores. Human ratings showed a broader distribution, while Claude produced a more compressed and optimistic scoring pattern. Despite this bias, the relative relationships between property scores and human judgments remained useful. Across the full dataset, Gradients emerged as the strongest overall predictor, suggesting that smooth transitions, layered visual change, and gradual shifts in spatial or textural structure are strongly associated with perceived scenic quality. For urban images, the strongest predictors were Simplicity and Inner Calm, Positive Space, Not-Separateness, Levels of Scale, and Gradients, indicating that attractive built environments were associated with coherence, spatial clarity, visual unity, and ordered complexity. For landscape images, the strongest predictors were Positive Space, Good Shape, Levels of Scale, Edge Density, and Simplicity and Inner Calm, suggesting that natural scenes were evaluated through both higher-order compositional qualities and measurable structural features. The addition of SAM-derived features improved R^2 only modestly, with the largest gain in landscape images, showing that segmentation-based structural information added value but did not replace the explanatory strength of Alexander-property ratings. These findings are consistent with the Results and Discussion sections of the current draft, which emphasize that Claude-derived properties dominated the predictive signal while SAM contributed secondary structural information.

The broader significance of this study lies in its contribution to computational aesthetics, explainable AI, architectural theory, and evidence-based design. For computational aesthetics, the study demonstrates that model outputs become more useful when they are interpretable. A single aesthetic score may predict attractiveness, but it does not explain what makes a place attractive. By contrast, a property-based approach can indicate whether scenicness is associated with calmness, positive space, visual integration, scale hierarchy, or gradual transitions. For architectural and landscape theory, the findings provide empirical support for selected components of Alexander's theory of wholeness, while also showing that not all properties contribute equally across contexts. For urban planning and landscape evaluation, the results suggest that visual quality can be discussed through measurable design-relevant features rather than only through subjective preference or expert intuition.

At the same time, several limitations should guide interpretation. Claude's property ratings showed a positive scoring bias and may have compressed variation in aesthetic judgment. The dataset relied on static Geograph UK images and crowd-sourced ratings, meaning that results may reflect photographic framing, weather, cultural context, and the limitations of image-based perception. Alexander's properties were operationalized indirectly through an LLM rather than through independent human coding or geometric measurement. SAM features were also limited to structural proxies that may not fully capture

architectural morphology, depth, materiality, or lived spatial experience. Finally, the analysis was correlational, so it cannot prove that changing a specific property would causally increase scenicness.

Future research should validate Alexander-property ratings using trained human coders, test the framework across broader cultural and geographic contexts, and examine causality through controlled image editing, generative design experiments, or virtual environments. More advanced spatial features, including façade parsing, depth estimation, isovist measures, and architecture-specific segmentation, could also improve the connection between computational features and lived environmental perception. Overall, this study advances interpretable aesthetic assessment by showing that human scenicness judgments are not only predictable, but partially explainable through identifiable visual properties. Its main contribution is to shift the question from whether a place appears beautiful to which visual structures help make it so.

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